

Gathering Structured Data on WRDS

[WRDS](#) is the platform from which you can download data for company and stock research.

- Try downloading test samples of data from [WRDS](#) to get a sense of how things work (see instructions further below).
 - Follow the instructions in the email from your lecturer to register for a WRDS account. (**Note:** The account might not be ready for use until a few working days later.)
- You are advised to use the following database to research US listed companies and their stocks:
 - **Fundamentals Annual** data - CRSP-Compustat Merged: <https://wrds-www.wharton.upenn.edu/pages/get-data/center-research-security-prices-crsp/annual-update/crspcompustat-merged/fundamentals-annual/>
- To download a test sample of (firm-level) [Fundamentals Annual](#) data,
 - Click the page link to access the web query form
 - Select Date range from **1990-01 to 1991-12**
 - Near the end of the section **Step 2**, Click the radio button for **Search the entire database**
 - In the section for **Screening Variables**, leave the default choices unchanged except for unchecking the box **CAD**
 - Leave the choices in **Step 3** unchanged
 - In the section for **Step 4**: Choose query variables,
 - leave the default choice (ie, Data Items) of the section for Select Variable Types unchanged
 - then scroll down to look for the **Search All** box (in grey) where you can enter characters to search for variables containing such characters.
 - For example, if you enter **Security**, you see the variable choice for **Security-level Identifier** to appear. Click on it to move the selection to Selected on the right-hand side.
 - Continue to search for and select all the variables shown on the Selected column in the following screenshot:

The screenshot shows the WRDS web query form interface. At the top, there are tabs for 'Search All' (24/981), 'Link Information' (7/8), 'Identifying Information' (4/7), and 'Identifying Information'. Below the tabs, there are two main columns: 'Select' and 'Selected'. The 'Select' column has a search box with 'prcc' entered and a list of variables: 'OPTPRCCA -- Options Cancelled - Price (OPTPRCCA)' and 'PRCC_C -- Price Close - Annual - Calendar (PRCC_C)'. The 'Selected' column has a 'Clear All' button and a list of 10 variables, all of which are checked with green checkmarks: 'Security-level Identifier (LIID)', 'First Effective Date of Link (LINKDT)', 'Last Effective Date of Link (LINKENDDT)', 'Primary Link Marker (LINKPRIM)', 'Link Type Code (LINKTYPE)', 'Historical CRSP PERMCO Link to COMPUSTAT Record (LPERMCO)', 'Historical CRSP PERMNO Link to COMPUSTAT Record (LPERMNO)', 'Company Name (CONM)', 'Ticker Symbol (TIC)', and 'CIK Number (CIK)'. The 'Selected' column also shows a count of '(24)'.

- Also search for and select the following variables: **fyear pddur upd fic adrr sich at sale oiadp ceq prcc_f csho capx gdwl** (**Note:** If you need to use other firm-level variables, add them to this list here.)
 - Note: You can click at the **Variable Descriptions** tab next to the **Query Form** tab on the web query form to find brief descriptions of variables (e.g., a variable's full name) and then go back to search by a variable's full name.
- In the “○ **AND** ○ **OR**” section for Conditional Statements (which will appear only after some variable has been selected), click the dropdown menu to select **Security-level identifier equal** and type **01**
- In the section for **Step 5: Select query output**, select
 - **comma-delimited text (*.csv)**
 - **Uncompressed**
 - **YYYY-MM-DD. (e.g. 1984-07-25)** (**Note:** This is the first on the list)
- Finally, enter your E-Mail Address and then click the **Submit Form** button. (You may also download the .csv file with compression as a .zip file. This is faster but require decompression after the download and may cause an issue for Mac users.)
- After you have downloaded the firm-level data .csv file,
 - rename it to your preference, e.g., **ccmfunda_01Jan1990_31Dec1991.csv**
- **Note:** .zip is a compression file format, which compresses files of other formats (such as .csv, .doc, .xlsx, pdf, etc) into a smaller size to facilitate a faster download or upload. Downloading data files first in .zip compressed files and then uncompressed them afterward is common because if you deal with several decades of data, the .csv data file can easily be over 100MB large whereas the .zip compressed file is only about 20MB large.
 - Python is not clever enough to automatically read the .csv data file inside the downloaded .zip compressed file. You need to decompress the downloaded .zip files to obtain the .csv data files inside.
 - That you can enter the .zip file as though it is a file folder and there you can open and read the .csv file **does not mean that it is not compressed**. You need to **move that .csv inside the .zip folder to outside of it** to decompress the .csv file from the compressed .zip folder.
 - Unfortunately, I don't have a Mac and cannot predict its behaviour for .zip files.